

Cryo-EM structure of the bicarbonate receptor GPR30

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Abstract

G-protein-coupled receptor 30 (GPR30) is a bicarbonate receptor that plays a vital role in cellular responses to extracellular pH and ion homeostasis. Despite its significance, the mechanisms by which GPR30 interacts with bicarbonate ions remain elusive. There is no consensus on a drug that targets GPR30, and the difficulty in the pharmacological analysis has limited biological and drug discovery researches on GPR30. Here, we present the cryo-electron microscopy structure of human GPR30 in the presence of bicarbonate ions at 3.2 Å resolution. Our structure reveals unique extracellular pockets and critical residues for bicarbonate binding and activation. Functional assays demonstrate that mutations in these residues impair bicarbonate-induced GPR30 activation, underscoring their importance in receptor function. This study also provides insights into the G-protein coupling, highlighting the structural divergence between GPR30 and other GPCRs. Our findings not only advance the understanding of the role of GPR30 in pH homeostasis but also pave the way for the development of high-affinity drugs targeting GPR30 for therapeutic interventions in diseases associated with acid-base imbalance.

eLife assessment

This study resolves a cryo-EM structure of the GPCR, human GPR30, which responds to bicarbonate and regulates cellular responses to pH and ion homeostasis. Understanding the ligand and the mechanism of activation is **important** to the field of receptor signaling and potentially facilitates drug development targeting this receptor. While the overall structures are **solid**, the identification of the bicarbonate binding site is only partly supported by the structural data and cell-based functional assays, leaving a major aim of the study **incomplete**.

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Introduction

The extracellular environment regulates cellular responses, as exemplified by pH and ion homeostasis. The acid-base balance, largely based on the bicarbonate buffer system *in vivo*, serves as a vital mechanism that maintains the optimal pH for cellular function.¹ Changes in the

extracellular pH and ion homeostasis are monitored by membrane channels and receptors, such as G protein-coupled receptors (GPCRs). However, it remains unknown whether and how the acid–base balance-related ions, protons and bicarbonate ions, bind to the receptors and cause conformational changes that lead to intracellular signal transduction.

Previous studies established four acid–base balance-related GPCRs^{2,3,4} that respond to an increase in proton concentration. None of the structures of these proton-sensing GPCRs in a proton-bound form has been determined, although the cryo-EM structures of apo-GPR134 (G2A) and its oxidized fatty acid- and N-acyl amino acid-bound forms have been reported.⁵ While these proton-sensing GPCRs are likely to be involved in pH homeostasis, particularly in the acidic tumor microenvironment, at inflamed sites, and in ischemia-reperfusion injury⁶, it remains unknown how protons cooperate with these receptors working as acid–base balance-related GPCRs.

We recently reported that the physiological concentration of bicarbonate ions, a counterpart of protons in the bicarbonate buffer system, activates G protein-coupled receptor 30 (GPR30), which leads to G_q-coupled calcium responses.⁷ Our study also demonstrated that GPR30 in brain mural cells regulates blood flow and ischemia–reperfusion injury. GPR30 was identified as a G protein-coupled estrogen receptor that mediates the rapid non-genomic action of estradiol (E2)⁸. However, despite numerous reports on the pleiotropic functions of GPR30 *in vivo*^{9,10}, there is controversy surrounding the responses of GPR30 to E2 *in vitro*¹¹, *ex vivo*¹², and *in vivo*¹³. The broad expression of Gpr30^{14,15}, including blood vessels^{16,17}, stomach, and lung, has also raised the possibility of its non-estrogenic functions. We demonstrated that three amino acids, E115, Q138, and H307, are essential for the bicarbonate-induced activation of GPR30, according to the public homology model (<https://gpcrdb.org/>)¹⁸. However, we did not examine whether bicarbonate ions interact with GPR30 and cause conformational changes. Moreover, to date, there is no consensus on a high-affinity drug that targets GPR30, and the difficulty in the pharmacological analysis has limited biological and drug discovery research on GPR30. To elucidate the bicarbonate–GPR30 interaction and to clarify the first structural identification of an acid–base balance-related GPCR, we report the cryo-electron microscopy (cryo-EM) structure of human GPR30 in the presence of bicarbonate ions.

Results

Overall structure

For the structural study, we used the full-length human GPR30 sequence. To efficiently purify the stable GPCR-G-protein complex, the receptor and mini-G_{sqi} were incorporated in a ‘Fusion-G system’ by combining two complex stabilization techniques^{19,20} (Figure 1—figure supplement 1A, B). The modified receptor and G-protein were co-expressed in HEK293 cells and purified by Flag affinity chromatography in the presence of 200 mM NaHCO₃. After an incubation with nanobody 35 (Nb35) and single chain scFv16, which binds to mini-G_{sqi}, the complex was purified by size exclusion chromatography (Figure 1—figure supplement 1C, D). The structure of the purified complex was determined by single-particle cryo-EM analysis, with an overall resolution of 3.21 Å (Table 1, Figure 1—figure supplement 2, and “Methods”). As the extracellular portion of the receptor was poorly resolved, we performed receptor focused refinement, yielding a density map with a nominal resolution of 3.30 Å, which was combined with the overall refined map. The resulting composite map allowed us to precisely build the atomic model of the components, including the receptor (residues 51 to 196, 207 to 288, and 296 to 340, G-proteins, and scFv16 (Figure 1A, B). Nb35 was not visible in the cryo-EM map, probably because of the effect of the mini-G_s modification, although the interactive residues with Nb35 were not changed.

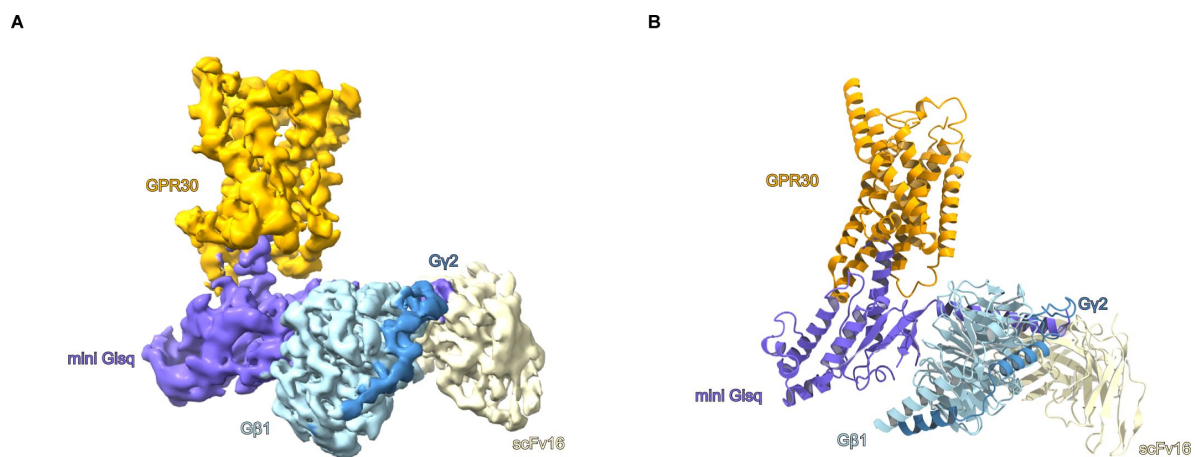


Figure 1.

Overall structure of the GPR30-miniG_{sqi}β₁Y₂-scFv16 complex.

A Unsharpened cryo-EM density map of the GPR30-miniG_{sqi}β₁Y₂-scFv16-Nb35 complex, with the components individually colored. **B** The refined structure of the complex is shown as a ribbon representation.

Table. 1

Cryo-EM data collection, refinement and validation statistics

Data collection	GPR30-G_q complex
Microscope	Titan Krios (Thermo Fisher Scientific)
Voltage (keV)	300
Electron exposure (e ⁻ /Å ²)	50.660
Detector	Gatan K3 Summit camera (Gatan)
Magnification	×105,000
Defocus range (μm)	-0.8–1.6
Pixel size (Å/pix)	0.83
Number of movies	9,824
Symmetry	C1
Picked particles	11,176,478
Final particles	148,285
Map resolution (Å)	3.21
FSC threshold	0.143
Model refinement	
Atoms	
R.m.s. deviations from ideal	
Bond lengths (Å)	0.004
Bond angles (°)	0.633
Validation	
Clashscore	9.17
Rotamers Outliers(%)	0.34
Ramachandran plot	
Favored (%)	93.77
Allowed (%)	6.23
Outlier (%)	0

The receptor consists of the canonical 7 transmembrane helices (TM) connected by three intracellular loops (ICL1–3) and three extracellular loops (ECL1–3), and the amphipathic helix 8 at the C-terminus (H8) (**Figure 2A**). Most of the TMs are kinked as in typical GPCRs²¹, but TM1 is also kinked at P71^{1.44} (superscripts indicate Ballesteros-Weinstein numbers²²), whereas TM1 adopts a straight α -helix in most class A GPCRs. ICL1 and ECL1 contain short α helices (**Figure 2B**, C). The short ICL3 was completely visible in the cryo-EM map (**Figure 2C** and **Figure 1**—figure supplement 3), while residues 197 to 206 in ECL2 and 289 to 295 in ECL3 were disordered (**Figure 2B**). ECL2 is attached to ECL1 by the disulfide bond between C130^{ECL1} and C207^{ECL2} (**Figure 2B**), which is highly conserved in class A GPCRs²¹. The cryo-EM structure did not superimpose well on the AlphaFold-predicted structure (Q99527 in the AlphaFold ^{19,20} database) with a root mean square deviation (R.M.S.D.) of C α atoms of 2.33 Å (**Figure 2C**—figure supplement 1A). The conserved D^{3.49}R^{3.50}Y^{3.51} motif²¹ in the predicted structure represents an inactive state (**Figure 2C**—figure supplement 1B). Moreover, ECLs 2 and 3 are rich in cysteine residues, which form incorrect disulfide bonds in the predicted structure (**Figure 2C**—figure supplement 1C). These comparisons support the usefulness of experimental structural determination.

Extracellular pockets

The interaction network between ECL1–3 covers the extracellular side of the receptor: ECL2 is anchored to ECL1 via the disulfide bond and ECL3 extends between ECL1 and ECL2, with Q296^{7.25} and F298^{7.27} located within the receptor cavity (**Figure 3A**). R299^{ECL3} and the backbone carbonyl oxygen of S297^{ECL3} interact with D210^{ECL2}, and N296^{7.25} forms polar interactions with E115^{2.60}, Q121^{2.66}, and N310^{7.39}. These interactions among the ECLs create four extracellular pockets (Pockets A–D) (**Figure 3B**). While Pockets A–C are positively-charged, Pocket D is negatively, and the size of each pocket is varied. Pocket A consists of ECL1, TM1, and TM7 (**Figure 3C**), while pocket B is formed by TM6, TM7, and ECL1 (**Figure 3D**). These two pockets are separated by R122^{ECL1} and H307^{7.36}. Pocket A is hydrophobic and appears unsuitable for bicarbonate binding (**Figure 3C**). Pocket B has hydrophilic residues (**Figure 3D**), but given the invisible region of ECL3 (residues 289 to 295), we cannot rule out the possibility that the pocket is buried. Pocket C consists of ECL2, ECL3, TM5, and TM6, and is positively charged due to the presence of arginine and histidine residues (**Figure 3B**, E). However, pocket C is relatively superficial compared to the other pockets. No density corresponding to bicarbonate is observed in these pockets (A–C), indicating that they are not ligand binding sites. Nonetheless, they may serve as potential binding sites for allosteric modulators or agonist drugs targeting GPR30.

Pocket D consists of ECL1–3 and TM2–7, and is the largest among the four pockets. Although the surface of pocket D is partitioned by the C130^{3.25}-C207^{45.52} disulfide bond and F208^{ECL2}, it is connected inside the receptor, deep enough to reach W272^{6.48}. Pocket D contains numerous hydrophilic residues such as aspartate, glutamate, asparagine, and glutamine, which favor bicarbonate binding, suggesting that pocket D is a good candidate for the bicarbonate binding site.

Insights into bicarbonate binding

To predict the bicarbonate binding site, we performed an exhaustive mutant analysis of hydrophilic residues in pockets A–D, which was not done in the previous study⁷. We confirmed the similar expression levels of the HA-tagged hGPR30 mutants by western blotting (**Figure 4**—figure supplement 1A, B). To evaluate the bicarbonate-induced activation of GPR30, we used the TGF α shedding assay²³. The HA-tagged GPR30 wild-type showed a bicarbonate-dependent response, while the H307^{7.36}A mutant completely abolished the response, consistent with previous findings⁷ (**Figure 4A**, B). These results indicate that the present experimental procedure is useful to evaluate the bicarbonate-induced activation of GPR30. Furthermore, the C207^{ECL2}A mutation that disrupts the disulfide bond between ECL2 and ECL1, and the P71^{1.44} mutant that

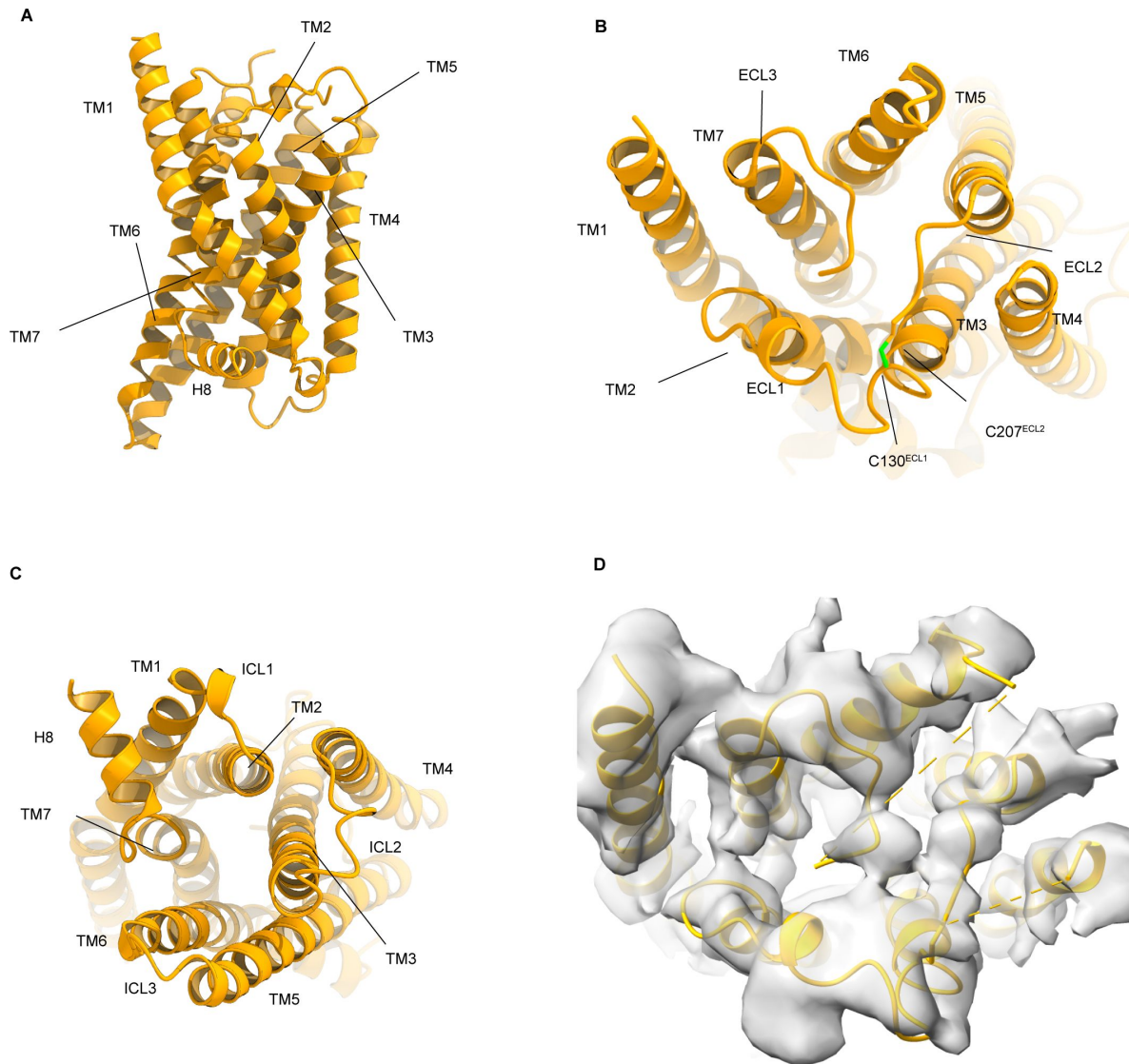


Figure 2.

Receptor structure.

A–C Overall structure of the receptor, viewed from the membrane plane (**A**), intracellular side (**B**), and extracellular side (**C**). **D** Extracellular view of the ribbon model overlaid the cryo-EM map.

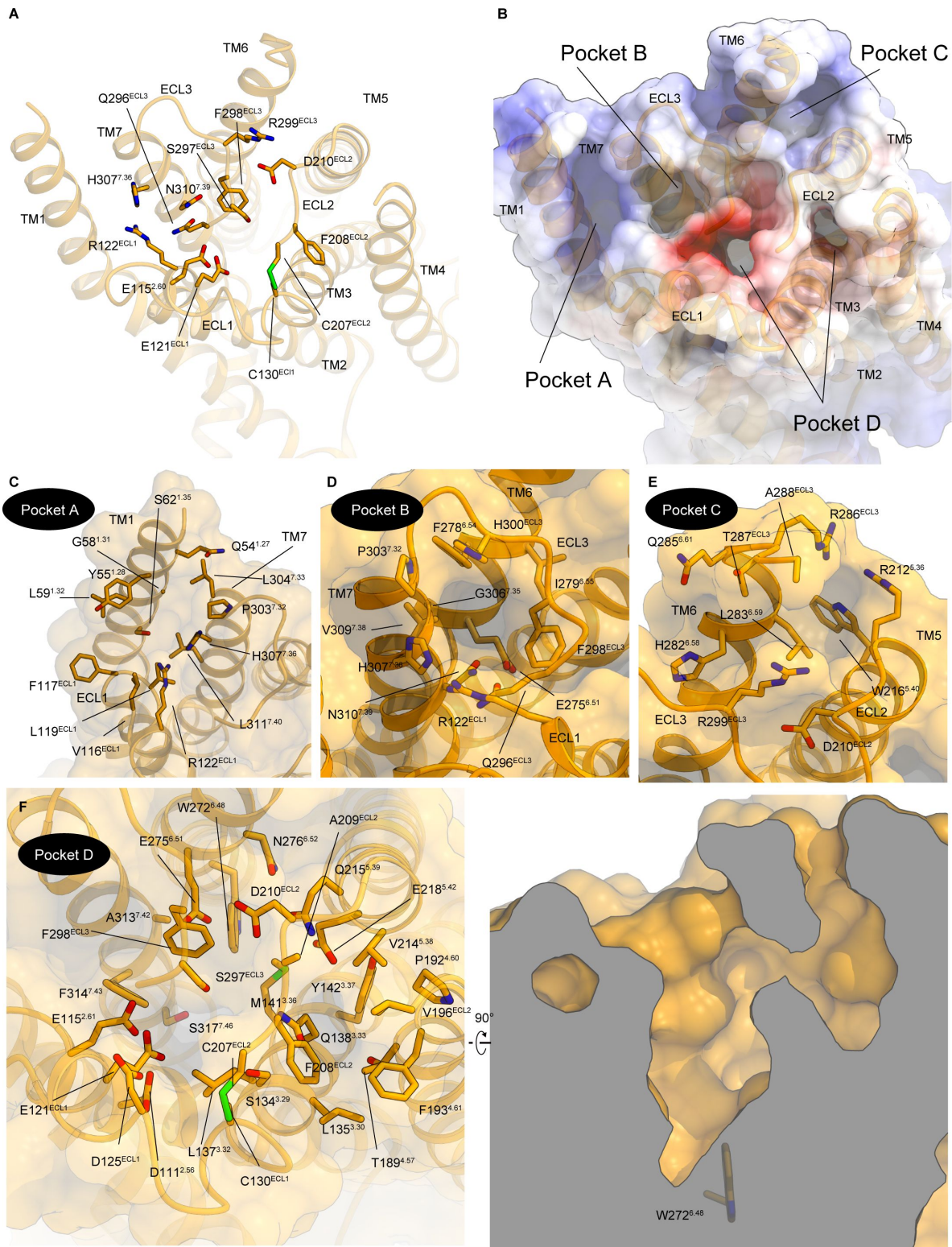


Figure 3.

Architecture of the extracellular pocket.

A Interactions between the ECLs. Black dashed lines indicate hydrogen bonds. **B** Molecular surface of the extracellular side. **C-F** Residues facing pocket A (**C**), pocket B (**D**), pocket C (**E**), and pocket D (**F**).

disrupts the characteristic kink in TM1, also abolished the bicarbonate response. These findings underscore the critical role of the structural integrity of the receptor, and provide compelling evidence that the bicarbonate response is mediated through GPR30.

Among the hydrophilic residue mutants, the alanine mutants of D125^{ECL1}, S134^{3.29}, D210^{ECL2}, and Q215^{5.39} reduced the bicarbonate response (**Figure 4A** [↗](#), **B**). Combining these data with previous studies, we mapped the residues essential for the bicarbonate response on the current structure (P71^{1.44}, E115^{2.60}, D125^{ECL1}, S134^{3.29}, Q138^{3.33}, C207^{ECL2}, D210^{ECL2}, Q215^{5.39}, and H307^{7.36}) (**Figure 4C** [↗](#)). Except for the structure-contributing residues (C207^{ECL2} and P71^{1.44}) and H307^{7.36}, the essential residues face pocket D, further supporting its role as the bicarbonate binding site. H307^{7.36} separates pockets A and B (**Figure 3C** [↗](#)), and the nearby mutants Q296^{7.25} and N310^{7.39} had no effect on the bicarbonate response, suggesting the indirect importance of stabilizing the active conformation of the receptor, rather than directly binding to bicarbonate.

Given the low affinity of GPR30 for bicarbonate⁷ [↗](#) and the challenges posed by carboxylic acid decarboxylation in cryo-EM analysis, visualizing bicarbonate at this resolution could be difficult. Nevertheless, we observed a weak density in pocket D, allowing us to assign bicarbonate (**Figure 4D** [↗](#)). The bicarbonate is surrounded by Q138^{3.33}, Q215^{5.39}, and E218^{5.42} (**Figure 4E** [↗](#)). Q138^{3.33} and Q215^{5.39} are residues with reduced bicarbonate responses due to mutations (**Figure 4C** [↗](#)). Most bicarbonate binding modes in other structures require recognition by asparagine and glutamine residues, in agreement with our model. Moreover, Q138^{3.33}, Q215^{5.39}, and E218^{5.42} are evolutionarily conserved in fish to human GPR30s (**Figure 4** [↗](#)—figure supplement 2). Below the bicarbonate binding site, M141^{3.36} forms a hydrophobic interaction with W272^{6.48} (**Figure 4D** [↗](#)), a toggle switch motif essential for receptor activation²¹ [↗](#), ²⁴ [↗](#). These observations allowed us to speculate that bicarbonate binding induces conformational changes in TM3 and TM5, facilitating receptor activation.

G-protein coupling

The C-terminal helix of Gα_q (α5-helix) is deeply enters the intracellular cavity of GPR30, resulting in the formation of an active signaling complex. On the C-terminal side of the α5-helix, the backbone carbonyl of Y243^{G.H5.23} (superscript indicates the common Ga numbering [CGN] system²⁵ [↗](#)) hydrogen bonds with R155^{3.50} in the conserved D^{3.49}R^{3.50}Y^{3.51} motif²¹ [↗](#) (**Figure 5A** [↗](#)). In addition, N244^{G.H5.24} forms hydrogen bonds with the peptide backbones of Y324^{7.53} and F331^{8.50}. ICL3 faces the middle part of the α5-helix, with three characteristic arginine residues (R248^{6.24}, R251^{6.27}, and R254^{6.30}) surrounding D233^{G.H5.13} (**Figure 5B** [↗](#)). R248^{6.24} and R254^{6.27} form hydrogen bonds with D233^{G.H.13} and Q237^{G.H5.17}, respectively. Moreover, R251^{6.30} extends toward the G-protein and hydrogen bonds with the peptide backbones of G207^{G.h4s6.11} and H209^{G.h4s6.13}. At ICL2, M163^{34.51} fits into a hydrophobic pocket composed of the α5-helix, the αN-β2 loop, and the β2-β3 loop of the Ga subunit, as in other GPCR-G_q complexes²⁶ [↗](#), ³⁰ [↗](#) (**Figure 5C** [↗](#), **H** [↗](#)). Above it, L159^{3.54} and A162^{34.50} form extensive hydrophobic interactions with L235^{G.H5.15}, L236^{G.H5.16}, and L240^{G.H5.20} in the α5-helix. These tight interactions with the α5-helix would enable the coupling of GPR30 with G_q.

The position of the α5-helix aligns with those in other GPCR-G_q complexes, but the position of the αN-helix is rotated away from the receptor (**Figure 5I** [↗](#)). The superimposition of the Ga subunits indicates that the rotation is not due to the movement of the entire G protein, but rather a 33° bend in the αN-helix (**Figure 5J** [↗](#)). This is attributed to variations in the interactions of the αN-helix with its receptors. The αN-helix interacts with the intracellular ends of ICL2 and TM4 in receptor-specific modes. In most G_q-coupled structures, ICL2 adopts an α-helix and R^{34.55} interacts with R38^{G.hns1.03} at the end of the αN-helix. A typical example is the head-to-toe interaction between R185^{34.55} and R32^{G.hns1.03} in the serotonin 5-HT_{2A} receptor²⁹ [↗](#) (**Figure 5E** [↗](#)). However, in GPR30, ICL2 does not adopt an α-helix and R^{34.55} is replaced by L167^{34.55}, which displaces R32^{G.hns1.03} downward through their interaction. Moreover, T170^{34.58} and R169^{34.57} form direct hydrogen bonds with R31^{G.hns1.02} and Y243^{G.H5.23}, respectively, which are not observed in other

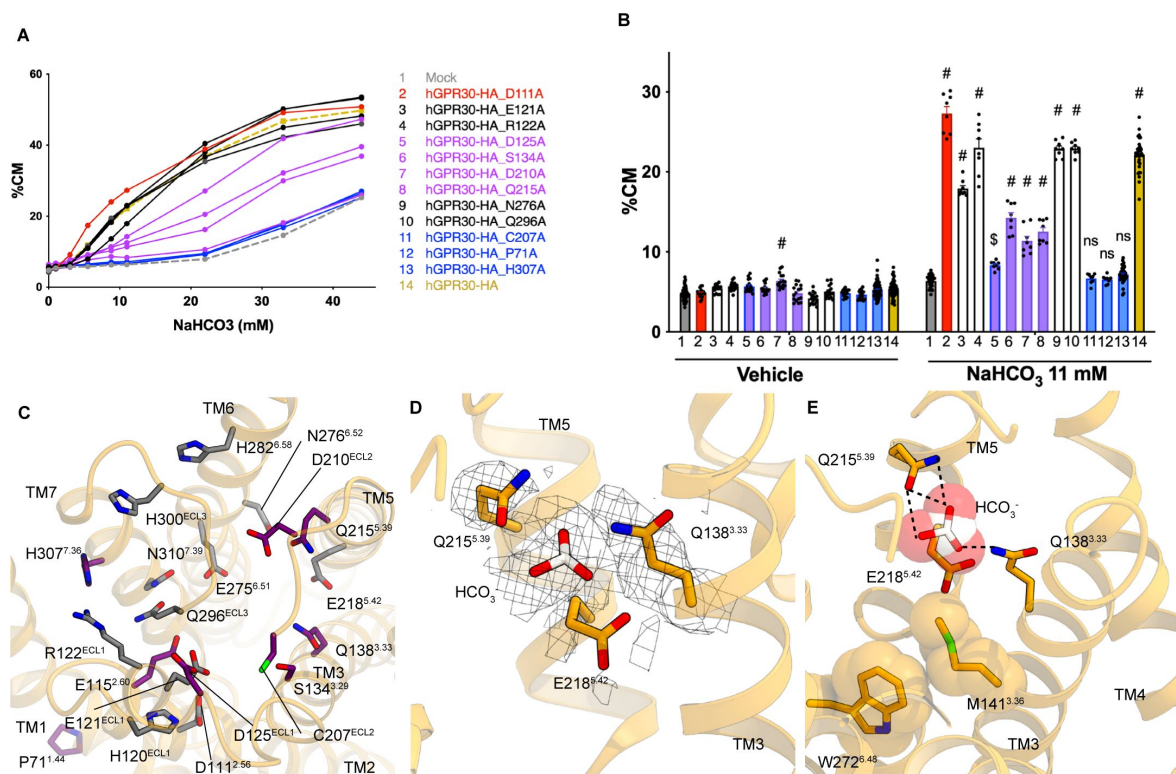


Figure 4.

Mutational analysis.

A, B TGF α shedding assay using HEK293 cells transfected with HA-tagged hGPR30. The mutants D111A; C207A, P71A, H307A; and D125A, S134A, D210A, Q215A are highlighted in red, blue, and purple, respectively. Statistical analysis: \$ $p = 0.0001$, # $p < 0.0001$ compared to mock cells using two-tailed unpaired t-test with Bonferroni's correction after two-way ANOVA. ns indicates no significant difference. Data are presented as mean values (**A**) and mean values $\pm$ SEM (**B**). **C** Residues that were subjected to mutant analysis are mapped to the structure. Only residues with reduced bicarbonate responses are colored in magenta. **D** Density corresponding to bicarbonate. **E** Putative binding mode of bicarbonate. Black dashed lines indicate hydrogen bonds.

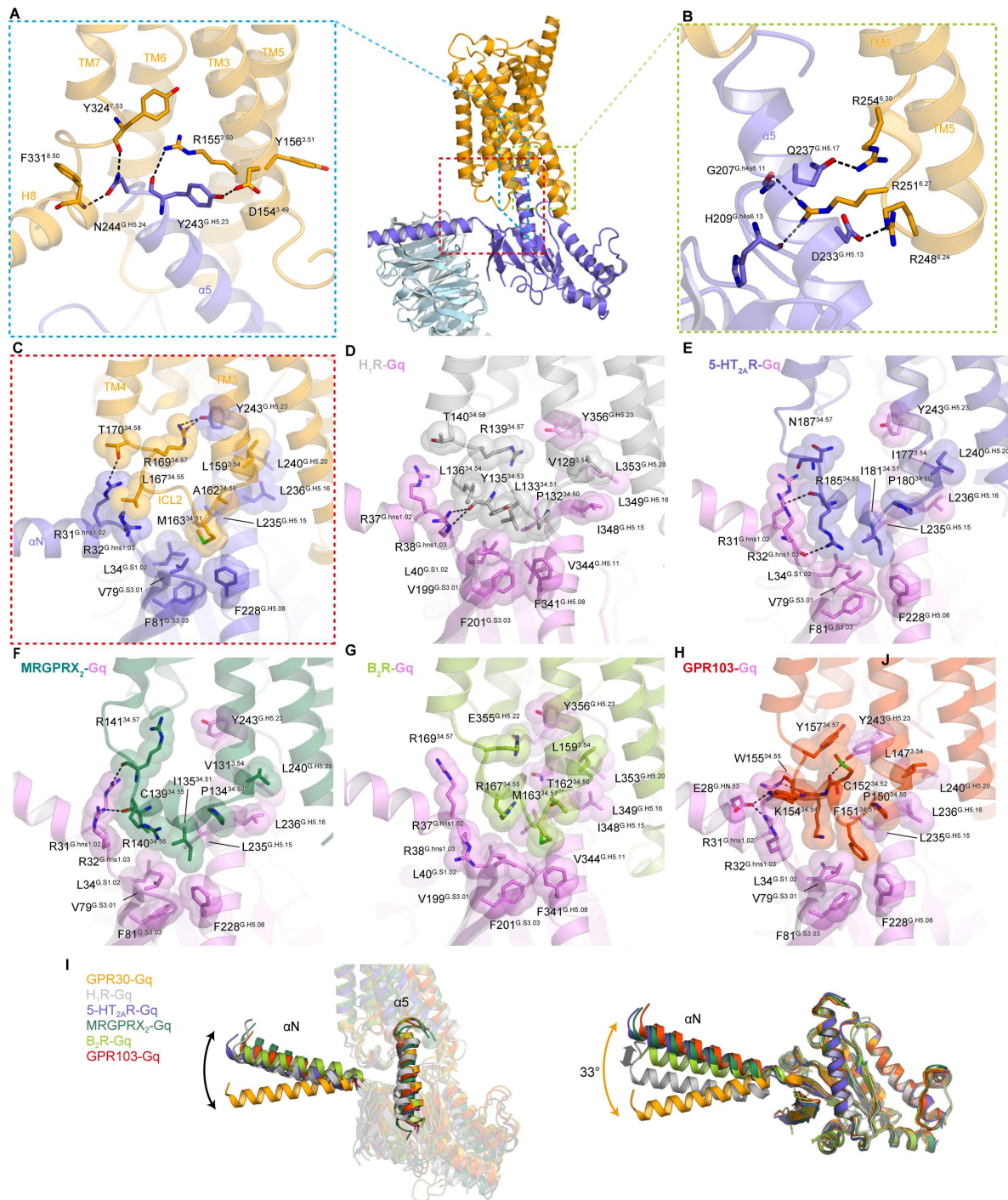


Figure 5.

G-protein coupling.

A Hydrogen-bonding interactions between the C-terminal $\alpha 5$ -helix residues and the receptor. **B** Electrostatic and hydrogen-binding interactions between ICL3 and the $\alpha 5$ -helix. **C-H** Interface between ICL2 and G_q , with residues involved in hydrophobic interactions represented by CPK models. G-protein-bound GPCRs used in the comparison are as follows: H_1R-G_q (PDB 7DFL, gray), B_2R-G_q (PDB 7F6I, yellow-green), $MRGPRX_2-G_q$ (PDB 7S8L, green), $5-HT_{2A}R-G_q$ (PDB 6WHA, gray), and $GPR103-G_q$ (PDB 8ZH8, red). **I** Comparison of the angles and positions of $\alpha 5h$ and αN relative to the receptor. **J** Superimposition of the G_a subunits.

GPCR-G_q complexes. We hypothesize that these interactions may underpin the distinct αN rotation: A similar interaction between R38^{G.hns1.03} and L136^{34.54} is also observed in histamine H1 receptor (H₁R)²⁶ (Figure 5D), in which translational rather than rotational motion of αN is observed (Figure 5J).

Discussion

Our study has elucidated the distinctive structure of GPR30, characterized by the kink of TM1 and the presence of multiple extracellular pockets formed by ECLs. Structural analysis and unbiased mutagenesis experiments on hydrophilic residues highlighted the largest pocket, designated as pocket D, as pivotal for bicarbonate recognition. Through putative bicarbonate modeling, we have proposed the binding mode of bicarbonate ions and the receptor activation mechanism. Overall, the elucidated structure of the extracellular domain of GPR30 and the insights into bicarbonate binding could facilitate the development of high-affinity drugs targeting GPR30.

It should be noted that our current model cannot fully explain all the mutagenesis results, such as those involving E115^{2.60}, D125^{ECL1}, and D210^{ECL2}. Although negatively charged residues generally repel bicarbonate, some structures have indicated their involvement in bicarbonate recognition. Thus, it is conceivable that these residues are transiently involved in bicarbonate binding, or two bicarbonate molecules might be accommodated within pocket D of GPR30. Future studies should focus on higher-resolution structures and approaches such as using bicarbonate derivatives to further evaluate bicarbonate binding.

GPR30 exhibits low homology to other GPCRs, with even closely related receptors sharing less than 30% sequence identity. A BLAST analysis revealed that GPR30 shares the highest sequence identity of 28% with the type 2 angiotensin II receptor (AT₂R) (Figure 6—figure supplement 1). Consequently, we compared the current structure of GPR30 with the ligand-bound AT₂R. The superimposition of GPR30 and AT₂R, with a Cα R.M.S.D. of 2.17 Å, suggests significant structural divergence between the two (Figure 6A). Notably, there is no structural commonality between GPR30 and AT₂R in terms of the orientations of their respective TMs. Specifically, the extracellular half of TM1 in GPR30 extends outward, and is kinked at P71^{1.44} in the center of TM1 (Figure 6B). Near P71^{1.44}, Y65^{1.38} and F70^{1.43} are oriented towards the interior of the receptor, forming hydrophobic interactions with L113^{2.58} and L331^{7.40} to stabilize the kink. In the examination of the loops, ECL2 of AT₂R features a short β-sheet commonly found in peptide-activated class A GPCRs, whereas that of GPR30 adopts an elongated conformation that covers the extracellular pocket essential for the bicarbonate response (Figure 4A, B). These comparisons underscore the unique structural characteristics of GPR30.

In most GPCRs, TM1 forms a straight α-helix and is positioned away from the orthosteric pocket formed by TM2–7. However, TM1 undergoes an inward movement due to allosteric conformational changes during agonist binding. The cannabinoid receptor CB₁ exemplifies this, with TM1 kinked at G127^{1.43} and folding inward upon ligand binding (Figure 6C). Similarly, the β₂ adrenergic receptor (β₂AR) and adenosine A_{2A} receptor (A_{2A}R) exhibit inward movements of TM1, with TM1 bending at the glycine residues rather than kinking (Figure 6D, E). By contrast, TM1 moves 4 Å outward in the endothelin ET_B receptor (Figure 6F). In GPR30, TM1 is kinked at P71^{1.44} (Figure 6B), indicating that this kink occurs regardless of the receptor activation state. Indeed, the P71^{1.44}A mutation, which eliminates the kink in TM1, completely abolished the bicarbonate response (Figure 4A, B). Moreover, P71^{1.44} is evolutionarily preserved (Figure 4—figure supplement 2). These findings suggest the critical role of the TM1 kink in both the putative conformational changes associated with ligand binding and the functional integrity of GPR30.

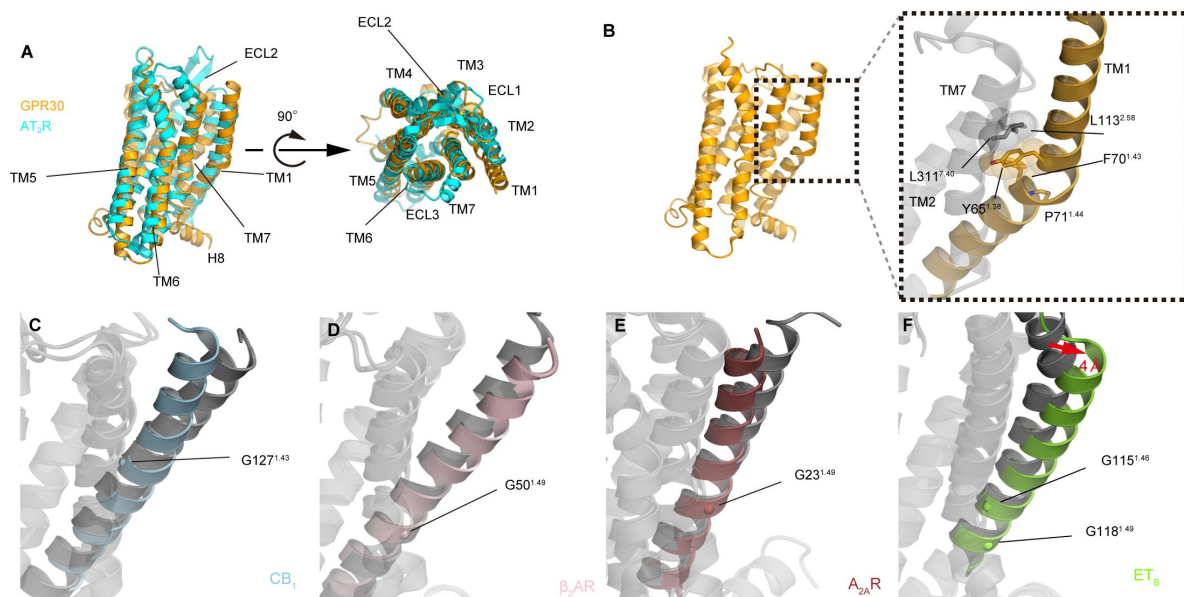


Figure 6.

Structural comparison with related GPCRS.

A Structural comparison of GPR30 and AT₂R (PDB 5UNF). **B** Interactions around P70 in TM1 of GPR30. **C–F** Conformational changes of TM1 upon agonist binding in CB₁ (**C**), β₂AR (**D**), A_{2A}AR (**E**), and ET_B (**F**). The agonist-bound states are colored with the respective colors, while the inactive states are colored gray. The PDB codes used in this figure are CB₁-active (PDB 5XRA), CB₁-inactive (PDB 5TGZ), β₂AR-active (PDB 3SN6), β₂AR-inactive (PDB 2RH1), A_{2A}AR-active (PDB 6GDG), A_{2A}AR-inactive (PDB 3EML), ET_B-active (PDB 8IY5), and ET_B-inactive (PDB 5X93).

Data Availability

The cryo-EM density map and atomic coordinates for the LPA₁-G_i complex have been deposited in the Electron Microscopy Data Bank and the PDB, under the accession codes: EMD-XXXXX [<https://www.ebi.ac.uk/emdb/entry/EMD-XXXXX>], and PDB XXXX [<http://doi.org/10.2210/pdbXXXX/pdb>]. Source data are provided with this paper. All other data are available from the corresponding authors upon reasonable request.

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Author contributions

S.K. performed all the experiments involved in the structural determination, assisted by H.A. and H.S.O. W.S. designed the project and initially tried the expression of GPR30. A.J.-W. and T.Y. performed and oversaw the mutagenesis study. The manuscript was mainly prepared by S.K. and W.S., with assistance from O.N.

Competing interests

O.N. is a co-founder and scientific advisor for Curreio. All other authors declare no competing interests.

Methods

Expression and purification of scFv16 and Nb35

The gene encoding scFv16 was synthesized (GeneArt) and subcloned into a modified pFastBac vector⁴⁶, with the resulting construct encoding the GP67 secretion signal sequence at the N terminus, and a His8 tag followed by a TEV cleavage site at the C terminus. The His8-tagged scFv16 was expressed and secreted by Sf9 insect cells, as previously reported⁴⁴. The Sf9 cells were removed by centrifugation at 5,000g for 10 min, and the secreta-containing supernatant was combined with 5 mM CaCl₂, 1 mM NiCl₂, 20 mM HEPES (pH 8.0), and 150 mM NaCl. The supernatant was mixed with Ni Superflow resin (GE Healthcare Life Sciences) and stirred for 1 h at 4 °C. The collected resin was washed with buffer containing 20 mM HEPES (pH 8.0), 500 mM NaCl and 20 mM imidazole, and further washed with 10 column volumes of buffer containing 20 mM HEPES (pH 8.0), 500 mM NaCl and 20 mM imidazole. Next, the protein was eluted with 20 mM

Tris (pH 8.0), 500 mM NaCl and 400 mM imidazole. The eluted fraction was concentrated and loaded onto a Superdex200 10/300 Increase size-exclusion column, equilibrated in buffer containing 20 mM Tris (pH 8.0) and 150 mM NaCl. Peak fractions were pooled, concentrated to 5 mg/ml using a centrifugal filter device (Millipore 10 kDa MW cutoff), and frozen in liquid nitrogen.

Nb35 was prepared as previously reported^{47,48}. In brief, Nb35 was expressed in the periplasm of *E. coli*. The harvested cells were disrupted by sonication. Nb35 was purified by nickel affinity chromatography, followed by gel-filtration chromatography, and frozen in liquid nitrogen.

Constructs for expression of GPR30 and Gq

The human GPR30 gene (UniProtKB, Q99527) was subcloned into a modified pFastBac vector⁴⁹, with an N-terminal haemagglutinin signal peptide followed by the Flag-tag epitope (DYKDDDDK) and the LgBiT fused to its C-terminus followed by a 3 C protease site and EGFP-His8 tag. A 15 amino sequence of GGSGGGSGSSSGG was inserted into both the N-terminal and C-terminal sides of LgBiT. Rat Gβ1 and bovine Gy2 were subcloned into the pFastBac Dual vector. In detail, rat Gβ1 was cloned with a C-terminal HiBiT connected with a 15 amino sequence of GGSGGGSGSSSGG. Moreover, mini-Gsqi was subcloned into the C-terminus of the bovine Gy2 with a nine amino sequence GSAGSAGSA linker. The resulting pFastBac dual vector can express the mini-Gsqi trimer.

Expression and purification of the human GPR30 – Gq complex

The recombinant baculovirus was prepared using the Bac-to-Bac baculovirus expression system (Thermo Fisher Scientific). For expression, 2 L of Sf9 cells at a density of 3×10^6 cells/mL were co-infected with baculovirus encoding GPR30 and miniGsqi trimer at the ratio of 1:1 and the cells were incubated at 30°C. After 48 hours, the collected cells were resuspended and dounce-homogenized in 20 mM Tris-HCl, pH 8.0(4°C), 200 mM NaCl, 10% Glycerol, 200 mM NaHCO₃, 25 mU/ml apyrase. The crude membrane fraction was collected by ultracentrifugation at 180,000g for 1 h and solubilized in buffer, containing 20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1% n-dodecyl-beta-D-maltopyranoside (DDM) (Calbiochem), 0.2 % cholesteryl hemisuccinate (CHS) (Merck), 10% glycerol, 200 mM NaHCO₃, and 25 mU/ml Apyrase, for 1 h at 4 °C. The supernatant was separated from the insoluble material by ultracentrifugation at 180,000g for 20 min and incubated with 5 mL of Anti-DYKDDDDK M2 resin (Sigma) for 1 h at 4°C. The resin was washed with 20 column volumes of buffer containing 20 mM Tris-HCl, pH 8.0 500 mM NaCl, 10% Glycerol, 0.05% Lauryl Maltose Neopentyl Glycol (LMNG) (Antrace), 0.005% CHS and 200 mM NaHCO₃. The complex was eluted in buffer containing 20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 10% Glycerol, 0.01% LMNG, 0.001% CHS, 200 mM NaHCO₃ and 0.2 mg/mL Flag peptide. The eluate was incubated with the Nb35 and scFv16 at 4 °C. The complex was concentrated and purified by size exclusion chromatography on a Superose 6 increase (GE) column in 20 mM Tris-HCl, pH8.0, 150 mM NaCl, 0.01% LMNG, 0.001% CHS and 200 mM NaHCO₃. Peak fractions were concentrated to 4.72 mg/ml for electron microscopy studies.

Sample vitrification and cryo-EM data acquisition

The purified complex was applied onto a freshly glow-discharged Quantifoil UltraAu grid (R1.2/1.3, 300 mesh), and plunge-frozen in liquid ethane by using a Vitrobot Mark IV. Data collections were performed on a 300kV Titan Krios G3i microscope (Thermo Fisher Scientific) equipped with a BioQuantum K3 imaging filter and a K3 direct electron detector (Gatan).

A total of 9824 movies were acquired with a calibrated pixel size of 0.83 Å pix⁻¹ and with a defocus range of -0.8 to -1.6 μm, using EPU. Each movie was acquired for 2.6 s and split into 48 frames, resulting in an accumulated exposure of about 50.660 electrons per Å².

Image processing

All acquired dose-fractionated movies were imported into CryoSPARC v4.4⁵⁰ and subjected to beam-induced motion correction. The contrast transfer function (CTF) parameters were estimated and a total of 10,148,422 particles were extracted. The particles were subjected to 2D classifications, Ab-initio reconstruction and several rounds of hetero refinement and Non-uniform refinement. Next, the particles were subjected to 3D classification with a mask on the receptor. Then the particle sets were subjected to Reference Based Motion Correction. Motion-corrected 148,285 particles were subjected to Non-uniform refinement, yielding a map with a global nominal resolution of 3.21 Å, with the gold standard Fourier Shell Correlation (FSC=0.143) criteria⁵¹. Moreover, the 3D model was refined with a mask on the receptor. As a result, the receptor has a higher resolution with a nominal resolution of 3.30 Å. The overall and receptor focused maps were combined by phenix⁵². The processing strategy is described in Supplementary Figure 2.

Model building and refinement

The density map was sharpened by phenix.auto_sharpen⁵³ and the quality of the density map was sufficient to build a model manually in COOT^{54,55}. The model building was facilitated by the Alphafold-predicted structure. We manually fitted GPR30, the Gq heterotrimer and scFv16 into the map. We then manually readjusted the model into using COOT and refined it using phenix.real_space_refine^{52,56} (v.1.19) with the secondary-structure restraints using phenix.secondary_structure_restraints.

Vector construction and transfection

Human *Gpr30* cDNA was obtained from human hepatoblastoma-derived HepG2 cells. The coding sequences of human *Gpr30* were inserted into the multi-cloning site of the plasmid vector pCXN2, which was generated in our laboratory via modification of pCAGGS, between the KpnI and EcoRI sites. The C-terminal HA-tagged *Gpr30* was amplified using a reverse primer containing the HA sequence. One amino acid mutation of human GPR30 was generated as follows: the targeted amino acid was changed to alanine (GCC) using a two-step PCR method with the QuikChange® Primer Design Program by Agilent (<https://www.agilent.com/store/primerDesignProgram.jsp>), and the coding sequence with each mutation was inserted into the multi-cloning site of pCXN2 between the KpnI and EcoRI sites. The mutations generated were D111A, E121A, R122A, D125A, S134A, D210A, Q215A, N276A, Q296A, C207A, P71A, and H307A.

Cell line sources and transfection

HEK293A cells (female origin; Thermo Fisher Scientific) were maintained at 37 °C and 5% CO₂. These vectors were transfected using the lipofection method (Lipofectamine™ 2000 Transfection Reagent, 11668019, Invitrogen).

TGFα shedding assay

The TGFα shedding assay was performed according to a previously published protocol²³. AP-TGFα expression vector was provided by Dr. Inoue and Dr. Aoki, Tohoku University. HEK293 cells were seeded in 12-well plates at a density of 1×10⁵ cells/well and cultured for 24 h. At 70% confluency, a mixture of plasmid vectors containing GPCR (see ‘**Vector construction and transfection**’ section) and AP-TGFα was transfected into the cells using Lipofectamine 2000 transfection reagent. After another 24 h of incubation, the cells were detached with 0.05% trypsin/EDTA (32777-44, Nacalai Tesque), suspended in Hanks’ balanced salt solution, and seeded in 96-well plates. The cells were stimulated for 1 h with 1–44×10⁻³ M (final concentration) of NaHCO₃ at 37 °C, under 0.03% CO₂. Conditioned media (CM) was transferred to another plate, and 80 µl of alkaline phosphatase (AP) solution (40 mM Tris-HCl, pH 9.5, 40 mM NaCl, 10 mM MgCl₂, 10 mM p-nitrophenylphosphate disodium salt hexahydrate) was added to both plates, which were

then incubated at 37 °C. The optical density at 405 nm (OD₄₀₅) was measured using a microplate reader (iMark, Bio-Rad) at 5 min, 30 min, 1 h, and 2 h, depending on the reaction rate. The percentage of shed AP-TGF was calculated using the following equations:

$$\text{AP activity} = \Delta\text{OD}_{405} (1 - 0 \text{ h})$$

$$\% \text{ CM (conditioned media)} = \text{AP}_{\text{CM}} / (\text{AP}_{\text{CM}} + \text{AP}_{\text{Cell}})$$

15 Analysis of cell surface expression of GPR30

The cell surface expression of wild-type and mutant hGPR30-HA was analysed via cell surface protein isolation using a Cell Surface Protein Isolation Kit (#89881, Thermo Scientific™) followed by western blotting. Briefly, HEK293 cells transiently expressing wild-type or mutant hGPR30-HA were biotinylated for 30 min at 4 °C. Then, the cells were harvested, and biotinylated proteins were isolated with avidin binding.

Western blotting

Cell lysates were separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis under reducing conditions and transferred to polyvinylidene difluoride membranes (Immobilon P, IPVH00010, Millipore). Primary antibodies used were anti-HA High Affinity, 1:1000 dilution, 11867423001, Roche; Na-K-ATPase, 1:1000 dilution, #3010, Cell Signaling Technology; β-Actin (AC-15), 1:1000 dilution, sc-69879, Santa Cruz Biotechnology. Secondary antibodies used were anti-rabbit IgG, HRP-linked Antibody, 1:5000 dilution, #7074, Cell Signaling Technology; anti-mouse IgG, HRP-linked Antibody, 1:5000 dilution, #7076, Cell Signaling Technology; anti-rat IgG, HRP-Linked Whole Ab Goat, 1:5000 dilution, NA935, Cytiva. The membranes were probed at 4 °C overnight with the primary antibodies. The membranes were subsequently incubated with the corresponding secondary antibodies. The signals were detected with ECL Prime (RPN2236, Cytiva) or ImmunoStar LD (296-69901, FUJIFILM Wako Pure Chemical Corporation) using a chemiluminescence imaging system (CFusion FX7, Vilber).

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Reviewer #1 (Public Review):

Summary:

This study resolves a cryo-EM structure of the GPCR, GPR30, which was recently identified as a bicarbonate receptor by the authors' lab. Understanding the ligand and the mechanism of activation is of fundamental importance to the field of receptor signaling. However, the main claim of the paper, the identification of the bicarbonate binding site, is only partly supported by the structural and functional data, leaving the study incomplete.

Strengths:

The overall structure, and proposed mechanism of G-protein coupling seem solid. The authors perform fairly extensive unbiased mutagenesis to identify a host of positions that are important to G-protein signaling. To my knowledge, bicarbonate is the only physiological ligand that has been identified for GPR30, making this study a particularly important contribution to the field.

Weaknesses:

Without higher resolution structures and/or additional experimental assessment of the binding pocket, the assignment of the bicarbonate remains highly speculative. The local resolution is especially poor in the ECL loop region where the ligand is proposed to bind (4.3 - 4.8 Å range). Of course, sometimes it is difficult to achieve high structural resolution, but in these cases, the assignment of ligands should be backed up by even more rigorous experimental validation.

The functional assay monitors activation of GPR30, and thus reports on not only bicarbonate binding, but also the integrity of the allosteric network that transduces the binding signal

across the membrane. Thus, disruption of bicarbonate signaling by mutagenesis of the putative coordinating residues does not necessarily mean that bicarbonate binding has been disrupted. Moreover, the mutagenesis was apparently done prior to structure determination, meaning that residues proposed to directly surround bicarbonate binding, such as E218, were not experimentally validated. Targeted mutagenesis based on the structure would strengthen the story.

Moreover, the proposed bicarbonate binding site is surprising in a chemical sense, as it is located within an acidic pocket. The authors cite several other structural studies to support the surprising observation of anionic bicarbonate surrounded by glutamate residues in an acidic pocket (references 31-34). However, it should be noted that in general, these other structures also possess a metal ion (sodium or calcium) and/or a basic sidechain (arginine or lysine) in the coordination sphere, forming a tight ion pair. Thus, the assigned bicarbonate binding site in GPR30 remains an anomaly in terms of the chemical properties of the proposed binding site.

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Reviewer #2 (Public Review):

Summary:

In this manuscript, "Cryo-EM structure of the bicarbonate receptor GPR30," the authors aimed to enrich our understanding of the role of GPR30 in pH homeostasis by combining structural analysis with a receptor function assay. This work is a natural development and extension of their previous work (PMID: 38413581). In the current body of work, they solved the first cryo-EM structure of the human GPR30-G-protein (mini-Gs α) complex in the presence of bicarbonate ions at 3.21 Å resolution. From the atomic model built based on this map, they observed the overall canonical architecture of class A GPCR and also identified 4 extracellular pockets created by extracellular loops (ECLs) (Pockets A-D). Based on the polarity, location, and charge of each pocket, the authors hypothesized that pocket D is a good candidate for the bicarbonate binding site. To verify their structural observation, on top of the 10 mutations they generated in the previous work, the authors introduced another 11 mutations to map out the essential residues for the bicarbonate response on hGPR30. In addition, the human GPR30-G-protein complex model also allowed the authors to untangle the G-protein coupling mechanism of this special class A GPCR that plays an important role in pH homeostasis.

Strengths:

As a continuation of their recent Nature Communication publication (PMID: 38413581), this study was carefully designed, and the authors used mutagenesis and functional studies to confirm their structural observations. This work provided high-resolution structural observations for the receptor in complex with G-protein, allowing us to explore its mechanism of action, and will further facilitate drug development targeting GPR30. There were 4 extracellular pockets created by ECLs (Pockets A-D). The authors were able to filter out 3 of them and identified that pocket D was a good candidate for the bicarbonate binding site based on the polarity, location, and charge of each pocket. From there, the authors identified the key residues on GPR30 for its interaction with the substrate, bicarbonate. Together with their previous work, they carefully mapped out nine amino acids that are critical for receptor reactivity.

Weaknesses:

It is unclear how novel the aspects presented in the new paper are compared to the most recent Nature Communications publication (PMID: 38413581). Some areas of the manuscript appear to be mixed with the previous publication. The work is still impactful to the field. The new and novel aspects of this manuscript could be better highlighted.

I also have some concerns about the TGF α shedding assay the authors used to verify their structural observation. I understand that this assay was also used in the authors' previous work published in Nature Communications. However, there are still several things in the current data that raised concerns:

(1) The authors confirmed the "similar expression levels of HA-tagged hGPR30" mutants by WB in Supplemental Figure 1A and B. However, compared to the hGPR30-HA (~6.5 when normalized to the housekeeping gene, Na-K-ATPase), several mutants of the key amino acids had much lower surface expression: S134A, D210A, C207A had ~50% reduction, D125A had ~30% reduction, and Q215A and P71A had ~20% reduction. This weakens the receptor reactivity measured by the TGF α shedding assay.

(2) In the previous work, the authors demonstrated that hGPR30 signals through the Gq signaling pathway and can trigger calcium mobilization. Given that calcium mobilization is a more direct measurement for the downstream signaling of hGPR30 than the TGF α shedding assay, pairing the mutagenesis study with the calcium assay will be a better functional validation to confirm the disruption of bicarbonate signaling.

(3) It was quite confusing for Figure 4B that all statistical analyses were done by comparing to the mock group. It would be clearer to compare the activity of the mutants to the wild-type cell line.

Additional concerns about the structural data include:

(1) E218 was in close contact with bicarbonate in Figure 4D. However, there is no functional validation for this observation. Including the mutagenesis study of this site in the cell-based functional assay will strengthen this structural observation.

(2) For the flow chart of the cryo-EM data processing in Supplemental data 2, the authors started with 10,148,422 particles after template picking, then had 441,348 Particles left after 2D classification/heterogeneous refinement, and finally ended with 148,600 particles for the local refinement for the final map. There seems to be a lot of heterogeneity in this purified sample. GPCRs usually have flexible and dynamic loop regions, which explains the poor resolution of the ECLs in this case. Thus, a solid cell-based functional validation is a must to assign the bicarbonate binding pocket to support their hypothesis.

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Reviewer #3 (Public Review):

Summary:

GPR30 responds to bicarbonate and regulates cellular responses to pH and ion homeostasis. However, it remains unclear how GPR30 recognizes bicarbonate ions. This paper presents the cryo-EM structure of GPR30 bound to a chimeric mini-Gq in the presence of bicarbonate. The structure together with functional studies aims to provide mechanistic insights into bicarbonate recognition and G protein coupling.

Strengths:

The authors performed comprehensive mutagenesis studies to map the possible binding site of bicarbonate.

Weaknesses:

Owing to the poor resolution of the structure, some structural findings may be overclaimed.

Based on EM maps shown in Figure 1a and Figure Supplement 2, densities for side chains in the receptor particularly in ECLs (around 4 Å) are poorly defined. At this resolution, it is unlikely to observe a disulfide bond (C130ECL1-C207ECL2) and bicarbonate ions. Moreover, the disulfide between ECL1 and ECL2 has not been observed in other GPCRs and the published structure of GPR30 (PMID: 38744981). The density of this disulfide bond could be noise.

The authors observed a weak density in pocket D, which is accounted for by the bicarbonate ions. This ion is mainly coordinated by Q215 and Q138. However, the Q215A mutation only reduced but not completely abolished bicarbonate response, and the author did not present the data of Q138A mutation. Therefore, Q215 and Q138 could not be bicarbonate binding sites. While H307A completely abolished bicarbonate response, the authors proposed that this residue plays a structural role. Nevertheless, based on the structure, H307 is exposed and may be involved in binding bicarbonate. The assignment of bicarbonate in the structure is not supported by the data.

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